DRUG STUDY

GENERIC NAME:
Cefoxitin

BRAND NAME: Monowel

MECHANISM OF ACTION:
Inhibits bacterial cell wall synthesis → bactericidal → effective against gram-positive and gram-negative bacteria, including anaerobes.

DRUG ILLUSTRATION:

INDICATION:

- Surgical prophylaxis
- Intra-abdominal infections
- Pelvic infections
- Post-operative infections

CLASSIFICATION:
2nd Generation Cephalosporin
Beta-lactam antibiotic

DOSE/ROUTE/FREQUENCY:
1g IV q8 (4AM, 12PM, 8PM)

CONTRAINDICATION:

- Allergy to cephalosporins
- Severe penicillin allergy

SIDE EFFECTS:

Common:

- Injection site pain
- Diarrhea
- Nausea

Serious:

- Anaphylaxis
- C. difficile-associated diarrhea
- Thrombophlebitis

NURSING RESPONSIBILITIES:

- Check allergy to penicillin/cephalosporins
- Monitor for anaphylaxis
- Observe IV site for phlebitis
- Monitor WBC count
- Complete full course

REFERENCES: Vallerand, A. H., Sanoski, C. A., & Quiring, C. (2023). *Davis's drug guide for nurses* (18th ed.). F.A. Davis Company.

DRUG STUDY

GENERIC NAME:
Ursodeoxycholic Acid (UDCA)

BRAND NAME: Euliv

MECHANISM OF ACTION:
Reduces cholesterol secretion into bile → decreases cholesterol absorption → gradually dissolves cholesterol gallstones → improves bile flow.

DRUG ILLUSTRATION:

INDICATION:

- Cholesterol gallstones
- Primary biliary cholangitis
- Post-cholecystectomy bile support

CLASSIFICATION:
Bile acid
Gallstone dissolving agent

DOSE/ROUTE/FREQUENCY:
1 tablet, 300 mg, PO OD (8 AM)

CONTRAINDICATION:

- Complete biliary obstruction
- Acute cholecystitis
- Hypersensitivity

SIDE EFFECTS:

Common:

- Diarrhea
- Abdominal discomfort

Rare:

- Elevated liver enzymes

NURSING RESPONSIBILITIES:

- Administer with food
- Monitor liver function tests
- Assess abdominal pain
- Educate patient that effect takes months

REFERENCES: Katzung, B. G., & Trevor, A. J. (2021). *Basic & Clinical Pharmacology* (15th ed.). McGraw-Hill Education.

DRUG STUDY

GENERIC NAME: Atorvastatin

BRAND NAME: Lipitor

MECHANISM OF ACTION:
Inhibits HMG-CoA reductase → decreases cholesterol synthesis in the liver → increases LDL receptor activity → lowers LDL, total cholesterol, triglycerides; slightly increases HDL.

DRUG ILLUSTRATION:

INDICATION:

- Hypercholesterolemia
- Prevention of cardiovascular disease
- Atherosclerosis
- Post-MI / high-risk cardiac patients

CLASSIFICATION:
HMG-CoA reductase inhibitor
Antihyperlipidemic (Statin)

DOSE/ROUTE/FREQUENCY:
1 tablet, 20 mg, PO OD (8 PM)

CONTRAINDICATION:

- Active liver disease
- Pregnancy & breastfeeding
- Hypersensitivity

SIDE EFFECTS:

Common:

- Headache
- Abdominal pain
- Myalgia

Serious:

- Rhabdomyolysis
- Hepatotoxicity

NURSING RESPONSIBILITIES:

- Check baseline lipid profile
- Monitor liver function tests (AST/ALT)
- Assess for muscle pain or weakness
- Give in the evening (cholesterol synthesis highest at night)
- Avoid grapefruit juice

REFERENCES: Brunton, L. L., Hilal-Dandan, R., & Knollmann, B. C. (2023). *Goodman & Gilman's The Pharmacological Basis of Therapeutics* (14th ed.). McGraw-Hill Education.

GENERIC NAME: Spironolactone

BRAND NAME: Aldactone

MECHANISM OF ACTION:
Blocks aldosterone receptors in distal renal tubules, promoting sodium and water excretion while retaining potassium.

DRUG ILLUSTRATION:

INDICATION:

- Heart failure
- Hypertension
- Edema
- Hyperaldosteronism

CLASSIFICATION:
Potassium-sparing diuretic; Aldosterone antagonist

DOSE/ROUTE/FREQUENCY:
25–100 mg PO daily (depends on condition)

CONTRAINDICATION:

- Hyperkalemia
- Severe renal failure
- Addison's disease

SIDE EFFECTS:

- Hyperkalemia
- Gynecomastia
- Menstrual irregularities
- Dizziness
- GI upset

NURSING RESPONSIBILITIES:

- Monitor potassium levels
- Monitor BP
- Assess for signs of hyperkalemia (muscle weakness, arrhythmia)
- Monitor I&O
- Avoid potassium supplements

REFERENCES: Brunton, L. L., Hilal-Dandan, R., & Knollmann, B. C. (2023). *Goodman & Gilman's The pharmacological basis of therapeutics* (14th ed.). McGraw-Hill.
Vallerand et al., 2023.

GENERIC NAME: Ceftazidime

BRAND NAME: Crizidime

MECHANISM OF ACTION:
Inhibits bacterial cell wall synthesis leading to cell death (bactericidal).

DRUG ILLUSTRATION:

INDICATION:

- Pneumonia
- UTI
- Sepsis
- Skin infections
- Pseudomonas infections

SIDE EFFECTS:

- Diarrhea
- Rash
- Allergic reaction
- Phlebitis (IV site)
- Superinfection

NURSING RESPONSIBILITIES:

- Check for penicillin/cephalosporin allergy
- Monitor WBC
- Monitor IV site for phlebitis
- Observe for anaphylaxis
- Encourage completion of antibiotic therapy

CLASSIFICATION: 3rd generation cephalosporin antibiotic

DOSE/ROUTE/FREQUENCY:
1–2 g IV every 8–12 hours (depends on infection severity)

CONTRAINDICATION:

- Hypersensitivity to cephalosporins
- Severe penicillin allergy (use cautiously)

REFERENCES: Brunton, L. L., Hilal-Dandan, R., & Knollmann, B. C. (2023). *Goodman & Gilman's The pharmacological basis of therapeutics* (14th ed.). McGraw-Hill.
Vallerand et al., 2023.

<table>
<tr><td>GENERIC NAME: Furosemide</td><td rowspan="2">MECHANISM OF ACTION:
Inhibits sodium and chloride reabsorption in the ascending loop of Henle, causing increased urine output.</td><td rowspan="5">SIDE EFFECTS:

● Hypokalemia
● Dehydration
● Hypotension
● Ototoxicity (high doses)</td><td rowspan="5">NURSING RESPONSIBILITIES:

● Monitor BP
● Monitor electrolytes (especially potassium)
● Strict I&O
● Daily weight
● Assess for dehydration
● Give in the morning to prevent nocturia</td></tr>
<tr><td>BRAND NAME: Lasix</td></tr>
<tr><td>DRUG ILLUSTRATION:
</td><td>INDICATION:

● Edema
● Heart failure
● Hypertension
● Pulmonary edema</td></tr>
<tr><td>CLASSIFICATION: Loop Diuretic</td><td rowspan="2">CONTRAINDICATION:

● Anuria
● Severe electrolyte imbalance
● Hypersensitivity</td></tr>
<tr><td>DOSE/ROUTE/FREQUENCY:
20–80 mg PO or IV daily (adjusted per condition)</td></tr>
<tr><td colspan="4">REFERENCES: Vallerand, A. H., Sanoski, C. A., & Quiring, C. (2023). Davis’s drug guide for nurses (18th ed.). F.A. Davis.</td></tr>
</table>

ASSESSMENT	DIAGNOSIS	PLANNING			IMPLEMENTATION	EVALUATION
		OBJECTIVES OF CARE	INTERVENTIONS	RATIONALE		
Subjective: ● "Nahihirapan po akong huminga." ● Reports needing 2–3 pillows to sleep. **Objective**: ● Dyspnea on exertion ● Orthopnea ● Bipedal edema (+2) ● Crackles on lung auscultation ● History of hypertension	Excess Fluid Volume related to compromised regulatory mechanism secondary to acute decompensated heart failure as evidenced by bipedal edema, orthopnea, and dyspnea.	The patient will demonstrate reduced edema within 16 - 24 hours. The patient will maintain balanced intake and output. The patient will verbalize understanding of fluid and sodium restriction before discharge.	Monitor vital signs (especially BP) every 4 hours. Strict intake and output monitoring. Daily weight at the same time each day. Assess degree of edema. Auscultate lung sounds every shift. Administer prescribed diuretics (e.g., furosemide, spironolactone). Implement fluid restriction as ordered. Educate patient on low-sodium diet.	Detects worsening fluid overload and hypertension. Evaluates effectiveness of therapy. Most accurate indicator of fluid retention. Monitor progression of edema. Identifies pulmonary congestion early. Promotes excretion of excess fluid. Prevents further fluid accumulation. Reduces sodium-induced water retention.	Monitored VS regularly and documented findings. Strict I&O chart maintained. Diuretics administered as ordered. Patient educated on sodium and fluid restriction.	Edema reduced from +2 to +1. Urine output adequate. Patient verbalized understanding of dietary restrictions.

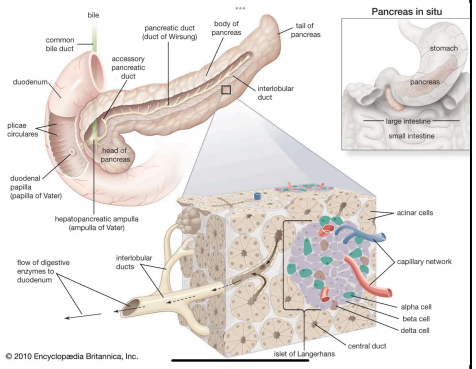
REFERENCES: NANDA International. (2021). *NANDA International nursing diagnoses: Definitions and classification 2021–2023*. Thieme.

ASSESSMENT	DIAGNOSIS	PLANNING			IMPLEMENTATION	EVALUATION
		OBJECTIVES OF CARE	INTERVENTIONS	RATIONALE		
Subjective: ● "Parang hindi ako makahinga." ● "Natatakot ako baka lumala pa ito." **Objective**: ● Restlessness ● Tachycardia ● Dyspnea ● Uses 2–3 pillows when sleeping (orthopnea) ● Facial tension / worried expression	Anxiety related to difficulty in breathing and hospitalization secondary to acute decompensated heart failure as evidenced by restlessness, verbalization of fear, and dyspnea.	The patient will verbalize reduced anxiety within 8 hours. The patient will demonstrate relaxation techniques. The patient will maintain stable vital signs within normal range.	Assess level of anxiety using verbal cues and behavior. Stay with the patient during episodes of dyspnea. Speak in a calm and reassuring manner. Encourage slow deep breathing exercises. Provide simple explanations about procedures and condition. Reduce environmental stimuli (noise, crowding). Administer oxygen therapy as ordered.	Determines severity and guides interventions. Provides reassurance and reduces fear. Helps decrease sympathetic response. Deep breathing reduces hyperventilation and promotes relaxation. Knowledge decreases fear of the unknown. Minimizes overstimulation that may worsen anxiety. Improves oxygenation and reduces sensation of breathlessness.	Assessed anxiety level regularly. Provided emotional support and remained with patient during dyspnea. Taught and assisted patient with deep breathing exercises. Maintained calm environment. Oxygen therapy administered as prescribed.	Patient verbalized feeling calmer. Restlessness decreased. Vital signs stabilized. Patient demonstrated proper breathing techniques.

REFERENCES: NANDA International. (2021). *NANDA International nursing diagnoses: Definitions and classification 2021–2023*. Thieme.

S T U D Y O F I L L N E S S C O N D I T I O N

Diabetes Mellitus type 2

ASSESSMENT	ANATOMY	PHYSIOLOGY	PATHOPHYSIOLOGY	ANALYSIS
Subjective: (Prior to Admission) • Polyuria • Polydipsia • Polyphagia • Fatigue • Blurred vision **Objective:** • Elevated FBS • Elevated RBS • HbA1c • Overweight • On oral hypoglycemics	The pancreas contains islets of Langerhans. **Beta cells** produce insulin, which acts on the liver, muscle, and adipose tissue to regulate blood glucose levels.  	After food intake, glucose enters bloodstream → pancreas releases insulin → insulin binds to receptors on cells → glucose transported into cells via GLUT4 transporters → used for energy or stored as glycogen in liver & muscle. Insulin also: • Inhibits gluconeogenesis • Promotes fat storage • Enhances protein synthesis Normal FBS: 70–99 mg/dL.	Insulin resistance develops in peripheral tissues → cells do not respond properly to insulin. The pancreas compensates by increasing insulin production, but over time beta-cell function declines → decreased insulin secretion → persistent hyperglycemia. Excess glucose causes osmotic diuresis (polyuria) and cellular starvation (polyphagia). Chronic hyperglycemia leads to endothelial damage, resulting in microvascular (retinopathy, nephropathy, neuropathy) and macrovascular (CAD, stroke) complications.	The patient's hyperglycemia results from insulin resistance and progressive beta-cell dysfunction. Chronic elevated glucose damages blood vessels, increasing risk for cardiovascular disease — especially significant because the patient also has hypertension. Strict glycemic control is essential to prevent complications. Management includes oral hypoglycemics, dietary control, exercise, and monitoring HbA1c. Nursing priority: monitor blood glucose, assess for hypo/hyperglycemia, promote lifestyle modification, and prevent long-term complications.

REFERENCES: Hinkle, J. L., & Cheever, K. H. (2022). *Brunner & Suddarth's textbook of medical-surgical nursing* (15th ed.). Wolters Kluwer.